Statistical Methods in Chemistry

Block 5

Analytical Method Validation

Student handout

Name: ______________________________

Class: ______________________________

Date: ______________________________

Contents

1 What method validation means

A chemistry method is not automatically reliable because it gives a number. A method must be tested to show that it is suitable for its intended purpose. This is the purpose of **analytical method validation**.

Core idea

Method validation is the process of collecting evidence that an analytical method gives results that are fit for a specific chemical purpose.

A validated method should answer questions such as:

Question	Meaning in chemistry
Can the method measure the correct substance?	Selectivity or specificity
Is the result close to the true value?	Accuracy, often tested by recovery or standards
Does the result repeat well?	Precision, usually expressed using standard deviation or RSD
Is the response proportional to concentration?	Linearity and calibration range
What is the smallest amount that can be detected?	Limit of detection, LOD
What is the smallest amount that can be measured reliably?	Limit of quantification, LOQ
Does the method tolerate small changes?	Robustness

Validation is not a single calculation. It is a structured judgement based on data, graphs, acceptance criteria, and chemical reasoning.

Validation is purpose-dependent

A method can be valid for one purpose but unsuitable for another. A colorimetric method may be good enough for a school investigation at $\mathrm{mmol\,dm^{-3}}$ concentrations, but not good enough for trace metal analysis at $\mathrm{\mu mol\,dm^{-3}}$ concentrations.

Quick exercise: Validation language

Match each validation term to the best description.

Term	Description
Accuracy	____________________
Precision	____________________
Linearity	____________________
LOD	____________________
Robustness	____________________

2 Planning a validation study

A validation study should begin with the intended use of the method. The same method can require different validation evidence depending on whether the aim is screening, teaching, quality control, or formal analysis.

Validation plan

A validation plan should state the analyte, matrix, concentration range, equipment, acceptance criteria, and the statistical tests or calculations that will be used.

Planning item	Example for copper sulfate colorimetry
Analyte	Cu^{2+} or copper sulfate concentration
Matrix	Aqueous solution prepared in the school laboratory
Range	$0.10\,\mathrm{mmol\,dm^{-3}}$–$0.80\,\mathrm{mmol\,dm^{-3}}$
Instrument	Colorimeter or spectrophotometer at selected wavelength
Blank	Distilled water or reagent blank
Acceptance criteria	$R^2 \geq 0.995$, recovery 95 %–105 %, repeatability RSD $\leq 2\,\%$
Decision rule	Accept only if all critical criteria are met

Quick exercise: Building acceptance criteria

You are validating an EDTA titration for calcium in mineral water. Suggest one acceptance criterion for each item.

a) Repeatability: ______________________________

b) Recovery from spiked sample: ______________________________

c) Concordant titres: ______________________________

d) Calibration or standardisation check: ______________________________

3 Precision: repeatability, intermediate precision, reproducibility

Precision describes how close repeated results are to each other. It does not prove that the result is correct; it only shows the level of scatter.

Levels of precision

Repeatability	Same analyst, same equipment, same day, same method.
Intermediate precision	Same laboratory, but different days, analysts, or instruments.
Reproducibility	Different laboratories. This is usually beyond a school practical, but the concept is important.

Precision is commonly reported using the relative standard deviation:

$$\mathrm{RSD}\,\% = \frac{s}{\bar{x}} \times 100\%.$$

Worked example: Repeatability of calcium results

A calcium method gives the following results in $\mathrm{mg\,L^{-1}}$:

42.6, 42.8, 42.7, 42.5, 42.7

The mean is $\bar{x} = 42.66\,\mathrm{mg\,L^{-1}}$, the sample standard deviation is $s = 0.114\,\mathrm{mg\,L^{-1}}$, and

$$\mathrm{RSD}\,\% = \frac{0.114}{42.66} \times 100 = 0.27\,\%.$$

If the acceptance criterion is RSD $\leq 2.0\,\%$, the repeatability passes.

Quick exercise: RSD decision

A titration method gives the following five titre values in cm^3:

$$24.60, \quad 24.65, \quad 24.62, \quad 24.66, \quad 24.61$$

Calculate the mean, sample standard deviation, and RSD. If the acceptance criterion is RSD $\leq 0.20\,\%$, does the method pass?

4 Accuracy and recovery

Accuracy is closeness to the true or accepted value. In method validation, accuracy is often tested using standards, certified reference materials, or spike recovery.

Recovery

A spike recovery test adds a known amount of analyte to a sample. The method should recover most of what was added.

$$\text{Recovery}\,\% = \frac{\text{measured increase}}{\text{amount added}} \times 100\%.$$

For a sample and spiked sample:

$$\text{Recovery}\,\% = \frac{C_{\text{spiked}} - C_{\text{sample}}}{C_{\text{added}}} \times 100\%.$$

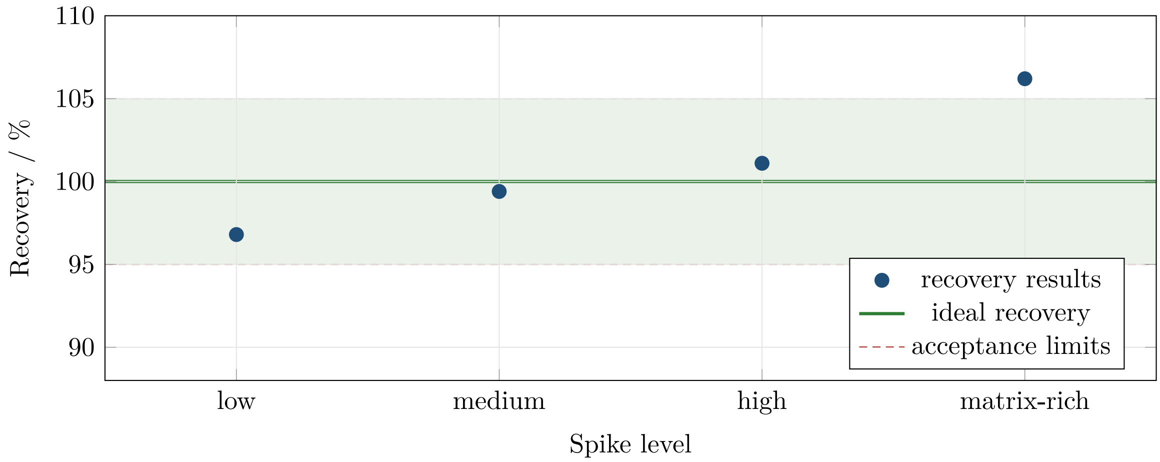

Figure 1. Spike recovery results. The shaded band shows a typical acceptance window of 95 %–105 %. The matrix-rich sample should be investigated because its recovery is above the limit.

Worked example: Spike recovery calculation

An unspiked sample contains 18.4 mg L^{-1} of analyte. A spike of 10.0 mg L^{-1} is added. The spiked sample is measured as 28.1 mg L^{-1}.

$$\text{Recovery}\,\% = \frac{28.1 - 18.4}{10.0} \times 100 = 97\,\%.$$

If the acceptance range is 95 %–105 %, this recovery passes.

Quick exercise: Recovery decision

An unspiked sample gives 0.320 mmol dm^{-3}. A 0.200 mmol dm^{-3} spike is added. The spiked sample gives 0.514 mmol dm^{-3}.

a) Calculate the recovery percentage.

b) Decide whether it passes an acceptance range of 95 %–105 %.

5 Linearity and working range

A method is linear when the measured response is proportional to analyte concentration over the stated range. A high R^2 value is useful, but it is not enough. The residuals and the chemistry must also make sense.

Linearity and range

The **linear range** is the concentration interval where the response is acceptably proportional to concentration. The **working range** is the interval where the method meets all required performance criteria: accuracy, precision, and acceptable uncertainty.

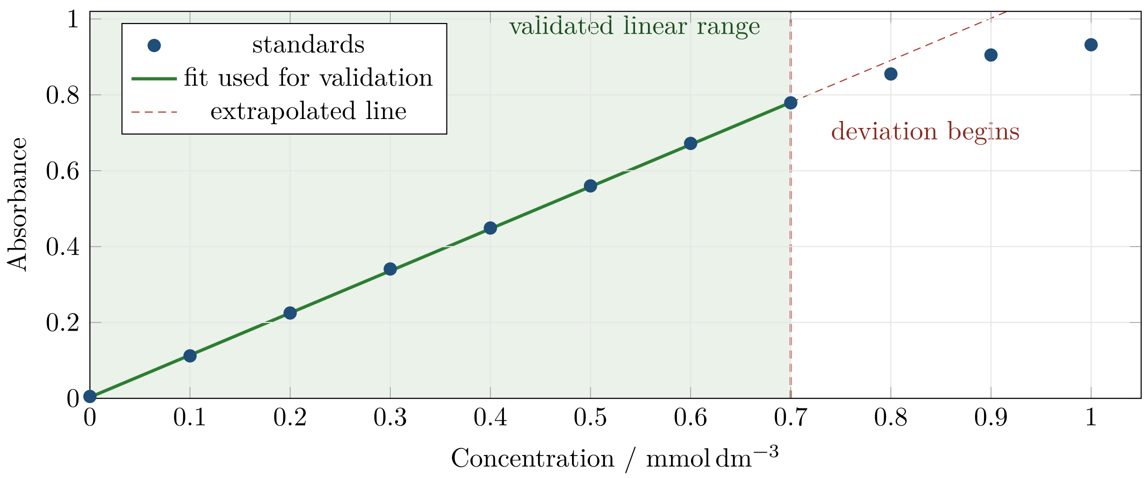

Figure 2. A calibration curve may look mostly linear but still deviate at high concentration. The fitted line should be based only on the validated range.

The residual plot is often more honest than the calibration plot.

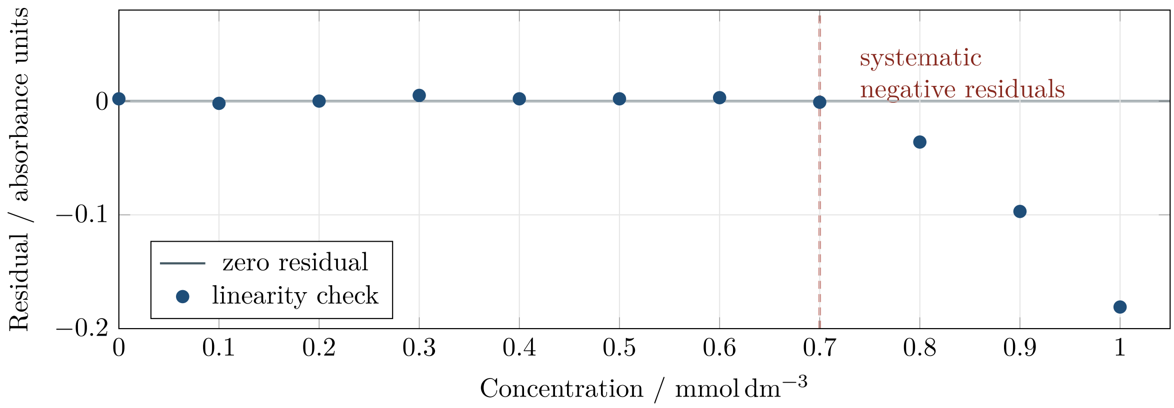

Figure 3. Residuals near zero in the low-to-middle range support linearity. A pattern of negative residuals at high concentration suggests curvature or detector saturation.

Quick exercise: Choosing a valid range

A colorimetric method has calibration points from 0.10 mmol dm^{-3}–1.00 mmol dm^{-3}. The residuals are random up to 0.70 mmol dm^{-3}, but become increasingly negative above this concentration.

a) What concentration range should be reported as validated?

b) Why should the highest standards not be used to calculate an unknown concentration?

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6 Sensitivity, LOD, and LOQ

Sensitivity describes how strongly the signal changes with concentration. In a linear calibration curve,

$$\text{sensitivity} = \text{slope of the calibration line.}$$

A steeper slope means a small concentration change produces a larger signal change.

Detection and quantification

LOD is the smallest concentration that can be distinguished from the blank. **LOQ** is the smallest concentration that can be measured with acceptable precision and accuracy.

A simple school-level estimate uses the standard deviation of blank measurements, s_{blank}, and the calibration slope, m:

$$\text{LOD} = \frac{3s_{\text{blank}}}{m}, \qquad \text{LOQ} = \frac{10s_{\text{blank}}}{m}.$$

Some formal protocols use $3.3s/m$ for LOD, but $3s/m$ is easier for teaching and gives the same practical idea.

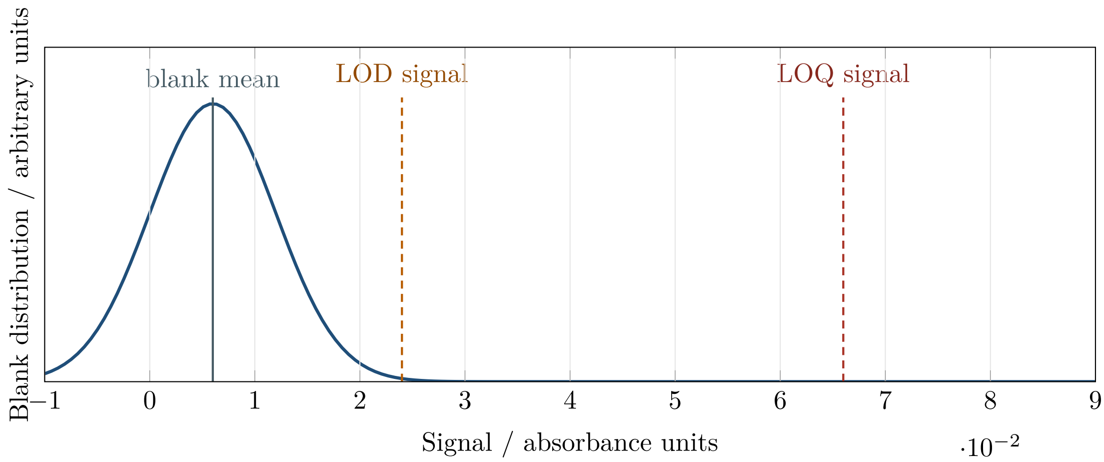

Figure 4. LOD and LOQ are based on how much signal is needed to rise above blank noise. Detection is not the same as reliable quantification.

Worked example: Calculating LOD and LOQ

Blank absorbance measurements have $s_{\text{blank}} = 0.0015$. The calibration slope is $m = 1.20$ absorbance mmol^{-1}dm^3.

$$\text{LOD} = \frac{3(0.0015)}{1.20} = 0.003\,75\,\text{mmol}\,\text{dm}^{-3}$$

$$\text{LOQ} = \frac{10(0.0015)}{1.20} = 0.0125\,\text{mmol}\,\text{dm}^{-3}.$$

A sample below the LOD should be reported as not detected. A sample between LOD and LOQ may be detected but should not be reported as accurately quantified.

Quick exercise: LOD and LOQ from blank data

Seven blank absorbance readings are:

$$0.006, \quad 0.008, \quad 0.004, \quad 0.005, \quad 0.007, \quad 0.006, \quad 0.005$$

The calibration slope is $m = 1.25$ absorbance mmol^{-1}dm^3.

a) Calculate the sample standard deviation of the blank readings.

b) Calculate the LOD.

c) Calculate the LOQ.

7 Selectivity and matrix effects

Selectivity is the ability of a method to measure the analyte in the presence of other substances. In real samples, the matrix can change the signal, endpoint, colour, pH, or reaction yield.

Matrix effect

A matrix effect occurs when other substances in the sample change the measured response of the analyte. The analyte may be present, but the method response is shifted by the sample background.

Examples in school chemistry:

- A coloured drink interferes with a colorimetric assay.
- Magnesium interferes with a calcium titration if the method is not selective.
- Turbidity scatters light and increases apparent absorbance.
- Buffer composition changes a complexometric titration endpoint.

A clean standard is not a real sample

A calibration curve made from pure standards does not automatically prove that the method works in juice, milk, soil extract, pond water, or mineral water. Matrix checks are part of validation.

Quick exercise: Identifying matrix effects

A copper colorimetry method works well for clear laboratory standards. The same method gives higher absorbance for a blue sports drink even after dilution.

a) What matrix effect may be present?

b) Suggest one control or correction.

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8 Robustness

Robustness tests whether small deliberate changes in the procedure cause unacceptable changes in the result. A robust method is not overly fragile.

Robustness

A method is robust if small, realistic changes in conditions do not significantly affect the final result.

Possible robustness checks:

Method	Possible robustness check
Titration	Slight pH change, endpoint delay, different indicator batch, different analyst
Colorimetry	Small wavelength change, cuvette orientation, blank preparation, waiting time
Reaction-rate method	Temperature variation, mixing time, stopwatch delay, reagent age
Gravimetry	Drying time, cooling time before weighing, filter type

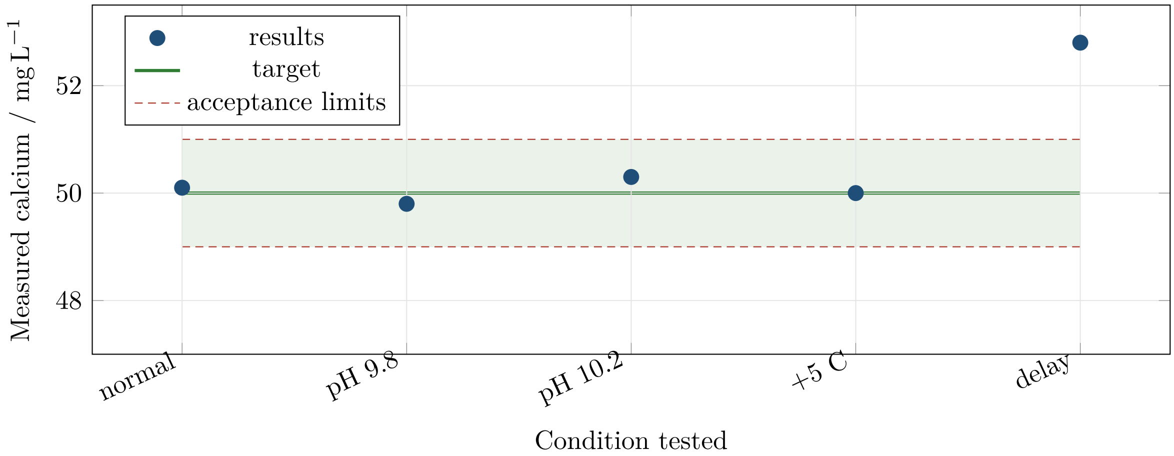

Figure 5. Robustness check for a calcium method. The endpoint delay condition falls outside the acceptance band and should be controlled in the SOP.

Quick exercise: Robustness decision

The target calcium value is $50.0\,\mathrm{mg\,L^{-1}}$. The acceptable range is $49.0\,\mathrm{mg\,L^{-1}}$–$51.0\,\mathrm{mg\,L^{-1}}$. In the robustness check above, which condition fails and what procedural instruction should be added?

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9 Reporting validation results

A validation report should be short, structured, and evidence-based. It should not simply say "the method worked." It should state which criteria were tested and whether they passed.

Minimum reporting structure

A useful validation report contains: method purpose, validation data, acceptance criteria, calculations, graphs, pass/fail decisions, limitations, and final scope of use.

Parameter	Evidence to report	Decision wording
Precision	Mean, standard deviation, RSD	RSD was below/above the criterion.
Accuracy	Recovery or standard result	Recovery was inside/outside the accepted range.
Linearity	Calibration equation, residuals, range	Linear over the stated range only.
LOD/LOQ	Blank SD, slope, calculated limits	Suitable/not suitable for the expected concentration.
Robustness	Effect of small method changes	Condition must/must not be tightly controlled.

Do not overclaim

If validation was only done for clear aqueous samples between $0.10\,\mathrm{mmol\,dm^{-3}}$–$0.80\,\mathrm{mmol\,dm^{-3}}$, do not claim that the method is valid for coloured drinks, soil extracts, or trace concentrations.

Quick exercise: Writing a defensible conclusion

Rewrite the weak conclusion into a stronger validation conclusion.
Weak conclusion: "The method is accurate and reliable because the graph had a high R^2."

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10 Mini-practical: validation summary

A student validates a copper colorimetry method for clear aqueous samples. The planned working range is 0.10 mmol dm^{-3}–0.80 mmol dm^{-3}. The following acceptance criteria are set before the experiment:

Parameter	Acceptance criterion
Repeatability	RSD $\leq 2.0\,\%$ at a mid-range concentration
Recovery	95 %–105 %
LOD	Less than 0.020 mmol dm^{-3}
Linearity	Residuals random within the stated range

Data

Repeatability results / mmol dm^{-3}	0.493	0.505	0.498	0.502	0.497

Recovery test	Result
Unspiked sample	0.320 mmol dm^{-3}
Spike added	0.200 mmol dm^{-3}
Measured spiked sample	0.514 mmol dm^{-3}

For the blank and calibration:

$$s_{\text{blank}} = 0.0025 \text{ absorbance units}, \qquad m = 1.10 \text{ absorbance mmol}^{-1}\text{dm}^3.$$

Quick exercise: Validation decision table

Complete the validation summary.

Parameter	Calculation or evidence	Pass/fail
Repeatability		
Recovery		
LOD		
Overall decision		

Quick exercise: Scope statement

Write one careful sentence describing what this method is validated for. Avoid overclaiming.

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11 Answer key

Validation language

Accuracy: closeness to the accepted or true value. Precision: closeness of repeated measurements to each other. Linearity: proportional response over a stated concentration range. LOD: smallest concentration that can be detected above blank noise. Robustness: resistance to small deliberate procedural changes.

Building acceptance criteria

Possible answers: repeatability RSD $\leq 2\,\%$; spike recovery $95\,\%$–$105\,\%$; at least three titres within $0.10\,\mathrm{cm}^3$; standardisation result within a stated tolerance such as $(0.1000 \pm 0.0020)\,\mathrm{mol\,dm^{-3}}$ or within $2\,\%$ of accepted value. Other defensible criteria are acceptable if they are clear and set before analysis.

RSD decision

Mean $= 24.628\,\mathrm{cm}^3$. Sample standard deviation $= 0.0259\,\mathrm{cm}^3$. RSD $= 0.0259/24.628 \times 100 = 0.105\,\%$. The method passes the RSD criterion of $0.20\,\%$.

Recovery decision

$$\text{Recovery}\,\% = \frac{0.514 - 0.320}{0.200} \times 100 = 97.0\,\%.$$

This passes the $95\,\%$–$105\,\%$ acceptance range.

Choosing a valid range

The validated range should be reported as approximately $0.10\,\mathrm{mmol\,dm^{-3}}$–$0.70\,\mathrm{mmol\,dm^{-3}}$. The highest standards should not be used because their negative residuals show systematic deviation from linearity; using them would bias unknown concentration estimates.

LOD and LOQ from blank data

For the blank readings, $s_{\text{blank}} \approx 0.00135$. With $m = 1.25$:

$$\text{LOD} = \frac{3(0.00135)}{1.25} = 0.0032\,\mathrm{mmol\,dm^{-3}}, \qquad \text{LOQ} = \frac{10(0.00135)}{1.25} = 0.0108\,\mathrm{mmol\,dm^{-3}}.$$

Identifying matrix effects

The drink may have its own colour or turbidity, causing background absorbance. Possible controls include a matrix-matched blank, standard addition, dilution into the validated range, filtration/centrifugation if turbidity is present, or measuring at a wavelength where interference is lower.

Robustness decision

The endpoint delay condition fails because $52.8\,\mathrm{mg\,L^{-1}}$ is outside $49.0\,\mathrm{mg\,L^{-1}}$–$51.0\,\mathrm{mg\,L^{-1}}$. The SOP should specify immediate endpoint reading or a maximum allowed delay after colour change.

Writing a defensible conclusion

Example: "The method is validated for clear aqueous samples within the tested concentration range because repeatability, recovery, and residual behaviour met the pre-set criteria. The method should not be used outside this range or for coloured/turbid matrices without additional validation."

Mini-practical

Repeatability data: mean $\bar{x} = 0.499\,\mathrm{mmol\,dm^{-3}}$, sample standard deviation $s \approx 0.0046\,\mathrm{mmol\,dm^{-3}}$, RSD $\approx 0.93\,\%$. This passes.

Recovery:

$$\frac{0.514 - 0.320}{0.200} \times 100 = 97.0\,\%.$$

This passes.

LOD:

$$\mathrm{LOD} = \frac{3(0.0025)}{1.10} = 0.0068\,\mathrm{mmol\,dm^{-3}}.$$

This is below $0.020\,\mathrm{mmol\,dm^{-3}}$, so it passes.

Overall decision: based on the supplied evidence, the method passes the stated criteria. A careful scope statement could be: "This colorimetric method is validated for determining copper concentration in clear aqueous samples within $0.10\,\mathrm{mmol\,dm^{-3}}$–$0.80\,\mathrm{mmol\,dm^{-3}}$ under the tested laboratory conditions."